

Overview & Outline

Purpose

This document is the research backbone for TransitSense. It is meant for the SIH team, mentors, and judges who need a clear view of what methods exist, what trade-offs they carry, and which ones are realistic for a student-built prototype.

This review supports three kinds of decisions:

1. Which methods are strong enough to cite in the pitch.
2. Which methods are practical for a low-cost, privacy-friendly, offline-tolerant MVP.
3. Which ideas belong in the bus demo now versus the train/metro extension story later.

Working principles

- Prefer peer-reviewed and technically credible sources.
- Judge every method on **accuracy, complexity, data needs, privacy, hardware cost, and connectivity assumptions**.
- Translate every paper into a **TransitSense fit** recommendation rather than just summarizing it.
- Keep the bus MVP as the main implementation focus, with train/metro as a scalable extension layer.

Outline

1. Overview

- Problem framing for public transport intelligence in the SIH context.
- Why buses are the deep prototype focus.
- What decisions this literature review should inform.

2. Bus tracking & ETA prediction

- GPS-based tracking pipelines.
- Map-matching and route alignment.
- Statistical, ML, and deep-learning ETA approaches.
- Handling sparse/noisy GPS and intermittent connectivity.
- Recommended MVP baseline and advanced option.

3. Passenger counting and density estimation

- Manual, sensor-based, and camera-based approaches.

- Occupancy classes vs exact counting.
- Edge inference and privacy-preserving analytics.
- Low-cost camera deployment constraints.
- Recommended MVP baseline and advanced option.

4. Station kiosks and real-time information displays

- Real-time arrival boards and freshness indicators.
- Offline-first kiosk behavior.
- Stop-level information design for passengers.
- Hardware assumptions for a student demo.

5. Route maps and wayfinding

- Bus route clarity and stop guidance.
- Metro and local-train line-map design patterns.
- Interchange communication and first-time-user support.
- What can be simulated well in a hackathon demo.

6. Privacy-preserving analytics and low-resource deployment

- On-device processing patterns.
- What data should never be stored.
- Edge/cloud split for poor-network conditions.
- Low-cost hardware trade-offs.

7. Comparison tables

- ETA methods comparison.
- Density methods comparison.
- Kiosk/wayfinding design comparison.
- Recommended default choices, stretch goals, and avoid-list.

8. Key decisions for TransitSense

- Recommended architecture choices.
- Risks and assumptions.
- Open research questions still worth exploring.

Update policy

As new research is added, each method summary should include:

- **Problem**

- **Data**
- **Method**
- **Results**
- **Constraints**
- **Fit for TransitSense**

This keeps the document usable as both a citation source and a design decision log.